

1. The following is the general equation to implement the Complementary Filter to perform sensor fusion using the readings obtained from the Accelerator and Gyroscope:

$$\Theta[n] = (1 - \alpha) * (\Theta_A[n]) + \alpha * (\Theta[n-1] + \omega_G[n] * \Delta t)$$

where

- Θ can be the Pitch or Roll angles derived based on sensor fusion
- Θ_A is the corresponding angle derived from the Accelerator's outputs values
- ω_G is the rate of change of the corresponding angle output by the gyroscope.

and α is the weightage associated with the Accelerator value and Gyroscope values.

Explain how the equation implements the sensor fusion function required by the operating principle of Complementary Filter, and discuss the limitation of this approach.

2. State the typical steps used in the implementation of Kalman Filter algorithm. Explain the purpose of each of these steps (in an intuitive manner), and how it can overcome the limitation of Complementary Filter.